

RegimeShift

SOQ 2026 Project (Advanced Track)
Performance Summary Report

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Date: July 25, 2026

Performance Summary

The proposed strategy combines Hidden Markov Models with convex portfolio optimisation to dynamically allocate capital across equity, gold and bonds. The portfolio is evaluated using an expanding walk-forward framework, ensuring that only historical information is used for training and optimisation. Performance is compared against two static benchmark portfolios.

Strategy	Sharpe	Sortino	Max DD	Calmar	Turnover
Dynamic Regime (HMM + CVXPY)	0.845	0.917	-19.01%	0.445	0.177
Equal Weight	1.482	1.800	-15.92%	0.751	—
60/40 Portfolio	0.838	1.012	-24.67%	0.334	—

The regime-based strategy performs competitively against the static benchmarks while adapting its allocation according to changing market conditions. Compared with the traditional 60/40 portfolio, it achieves a slightly higher Sharpe ratio, a lower maximum drawdown and a better Calmar ratio, indicating improved downside risk management.

Although the Equal Weight portfolio records the highest Sharpe and Sortino ratios over the evaluation period, the proposed strategy demonstrates the benefit of combining market regime detection with optimisation. By adjusting allocations across different market environments, it provides a balanced trade-off between return generation and risk control while maintaining moderate portfolio turnover.

Methodology

- Historical data for NIFTY, Gold ETF, LiquidBeES and India VIX were collected using `yfinance`.
- Market features including returns, momentum, volatility and VIX were used for regime identification.
- A three-state Gaussian Hidden Markov Model was trained using an expanding walk-forward framework.
- The detected market states were mapped to Bull, Bear and Crisis regimes.
- Convex optimisation using CVXPY determined the portfolio weights for each regime.
- Portfolio performance was evaluated after incorporating transaction costs and compared with static benchmark portfolios.

Conclusion

This project presents a systematic framework for regime-aware portfolio allocation using probabilistic market state detection and convex optimisation. The proposed approach successfully adapts portfolio weights across different market conditions while maintaining competitive risk-adjusted performance. The implementation follows a realistic walk-forward evaluation framework and demonstrates how machine learning and optimisation techniques can be combined for practical portfolio management.